

$$11) \sin 2x^2y^3 = 3x^3 + 1$$

$$12) 3x^2 + 3 = \ln 5xy^2$$

For each problem, use implicit differentiation to find $\frac{d^2y}{dx^2}$ in terms of x and y .

$$13) 4y^2 + 2 = 3x^2$$

$$14) 5 = 4x^2 + 5y^2$$

Critical thinking question:

- 15) Use three strategies to find $\frac{dy}{dx}$ in terms of x and y , where $\frac{3x^2}{4y} = x$. Strategy 1: Use implicit differentiation directly on the given equation. Strategy 2: Multiply both sides of the given equation by the denominator of the left side, then use implicit differentiation. Strategy 3: Solve for y , then differentiate. Do your three answers look the same? If not, how can you show that they are all correct answers?